Summary of Key Findings

- **Historical Financial Extremes:** The state utility (KSEB Ltd) has seen massive revenue gap rejections, with a peak rejection of **Rs. 2,498.47 crore** in FY 2016-17, primarily due to disallowances in O&M expenses. In contrast, small licensees like KPUPL have accumulated record surpluses, reaching a cumulative **Rs. 89.11 crore** by 2024.
- **Anomalous Approvals:** There are documented cases where the approved surplus exceeded the requested amount—such as KDHPCL in 2019-20 (requested Rs. 7.34 lakh, approved **Rs. 27.10 lakh**) and KPUPL in 2022-23, where the Commission determined higher non-tariff income from interest on surplus than the licensee had accounted for.
- **Regulatory Logic Patterns:** Staff analysis follows a strict hierarchy: disaggregating strategic business units, applying **normative caps** on O&M expenses (allowing only inflation-adjusted growth), and shared "efficiency gains" for distribution loss reductions. Certain items, like Section 3(1) duty and CSR, are perennially rejected.

Recommended Backend Architecture to Build Right Now

To move beyond simple chat interfaces and create a "brain" capable of replacing manual auditing, you should implement a **Neuro-Symbolic SLM Framework** (specifically utilizing the **LiR3AG** strategy). This architecture ensures mathematical precision that standard RAG systems lack.

1. Intelligent Ingestion Layer (Vision + NLP)

- **Function:** Automated extraction of complex financial tables from audited account PDFs.
- **Tech:** Use a multi-modal approach combining computer vision (like YOLOv5) for table structure recognition and NLP for semantic text extraction. This achieves up to 98% recognition accuracy, identifying the "Forms" and "Annexures" required by law.

2. Reasoning Orchestrator (LiR3AG Strategy)

- **Function:** Generating multiple reasoning chains to verify a single petition.
- **Logic:**
    - **Document Clustering:** Group retrieved regulations and past orders into "perspectives" (e.g., Technical, Financial, Legal).
    - **Parallel Drafting:** Use a **Small Language Model (SLM)** like Mistral-7B or Llama 3.1 (7B to 14B parameters) to generate parallel drafts of the truing up analysis.
    - **Domain Expertise:** Fine-tune the SLM on KSERC-specific terminology to ensure it understands nuanced concepts like "Net Fixed Assets" and "Regulatory Surplus".

3. Symbolic Logic Verifier (The "Guardrail")

- **Function:** The "Logic Brain" that provides a mathematical guarantee of correctness.
- **Implementation:** Instead of letting the AI "guess" the final number, pass the generated drafts through a **Symbolic Solver**.
- **Core Logic Rules:** Program the solver with formal mathematical constraints based on the 2021 Tariff Regulations:
    - `If Actual_O&M > Normative_O&M, then Approve = Normative_O&M`.
    - `Interest_on_Surplus = Accumulated_Surplus * SBI_Retail_Term_Deposit_Rate`.

4. On-Premise SLM Deployment

- **Function:** Data privacy and cost control.
- **Tech:** Host these compact models (SLMs) on your own GPU-enabled servers rather than using third-party clouds.

- **Benefit:** This keeps sensitive audited data inside the firewall and reduces operational costs by **99%** (e.g., Mistral-7B costs ~$15,000/year for massive scale compared to millions for GPT-4).

This hybrid approach—combining the language understanding of **SLMs** with the deterministic "Logic Guard" of a **Symbolic Solver**—creates a system that doesn't just generate text, but proves its results mathematically against regulatory code.